

**AMENDMENT NO. 1 JUNE 1999
TO
IS 2386 (PART VIII) : 1963 METHODS OF TEST
FOR AGGREGATES FOR CONCRETE**

PART VIII PETROGRAPHIC EXAMINATION

(Page 20, clause 3.6.2.1) — insert the following new clauses after 3.6.2.1:

3.7 Additional Test Method

3.7.1 Object

This test method describes procedure for determining undulatory extinction (UE) angle of coarse and fine aggregates and shall be conducted on aggregates containing generally more than 30 percent quartz. The undulatory extinction angle is the angle between the positions of extinction and its disappearance in the grain. It is a measure of the crystal lattices disturbance caused by geological stresses. Various researchers have used UE angle as indication of possible source for alkali reactivity of the rock.

3.7.2 Procedure

The undulatory extinction angle shall be determined in transmitted light (thin section) under a polarising microscope.

3.7.2.1 When the first extinction in the selected quartz grain showing the interference colour is visible, the position of graduated rotating stage of the polarising microscope shall be recorded.

3.7.2.2 Determine the second position by rotating the same grain with the rotating stage of the microscope through other extinction areas in the grains until the entire grain becomes bright coloured again except for some slightly grey shadows in the area covered 3.7.2.1. Determine the angle between the two positions.

3.7.2.3 Repeat the measurements mentioned in 3.7.2.1 and 3.7.2.2 in the reverse direction also. This will improve the precision of the test. the difference between the two measurements, that is forward and reverse directions, should be small.

3.7.2.4 Average the values (that is difference between the two positions) obtained in the two directions for a particular grain. This shall be the characteristic UE angle for a given grain.

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3.7.2.5 The angle characterising a thin section shall be obtained by averaging the values of all grains examined in the thin section.

3.7.2.6 The percentage of quartz grains showing undulatory extinction in the same section shall be determined by point-counting method.

NOTE — Some coarse aggregates may be of composite nature, that is, containing different rock types for example quartzite, granites, sandstone, etc. The undulatory extinction angle and the percentage of quartz grains showing UE shall be determined for each of the predominant constituents of the coarse aggregates.

(CED 2)